ACID PROPERTIES IN DATABASES

MODULE QUICK REVISION NOTES & INTERVIEW PREPARATION

1 What are ACID Properties?

Core Definition: ACID is a set of four foundational properties (Atomicity, Consistency, Isolation, Durability) that guarantee database transactions are processed reliably, preserving data integrity even under system failures, crashes, or high concurrency.



Property	Core Principle	Database Mechanism / Real-World Example
Atomicity	**"All or Nothing"** Entire transaction completes or entire transaction is rolled back.	Fund transfer from Account A to B: deducting money from A and adding to B must both succeed, or zero changes occur.
Consistency	**State Validity** Database transitions strictly from one valid schema state to another.	Enforces foreign keys, UNIQUE constraints, and CHECK rules. Account balance cannot drop below zero if constrained.
Isolation	**Concurrency Control** Concurrent execution yields identical results to serial execution.	Uses locks and Multi-Version Concurrency Control (MVCC) to prevent dirty reads and uncommitted interference.
Durability	**Permanence** Committed state changes survive system crashes and power failures.	Write-Ahead Logging (WAL) and non-volatile storage flushing ensure recovery after severe system crashes.

2 Summary of Benefits

Advantage	Technical Explanation
Data Integrity	Guarantees partially executed updates never corrupt state or leave orphaned records.
Crash Resilience	Provides immediate recovery vectors using transaction logs (undo/redo logs) upon reboot.
Safe Concurrency	Allows thousands of concurrent users without race conditions or read/write collisions.

3 Key Interview Questions & Answers

Q1: What is the exact distinction between Atomicity and Consistency?

Atomicity is an operational guarantee that a multi-step transaction is treated as a single indivisible unit (it either fully succeeds or completely rolls back). **Consistency** is an invariant guarantee that the database transitions from one structurally valid state to another, strictly satisfying all defined schema rules, keys, and checks. Atomicity is one of the underlying mechanisms used to achieve overall Consistency.

Q2: How do databases guarantee Atomicity and Durability during a crash?

Databases implement **Write-Ahead Logging (WAL)**. Before any data modification is written to disk tables, the intentional change is sequentially appended to a log file stored in non-volatile storage.

- **Undo Phase (Atomicity):** On recovery after a crash, uncommitted transactions are rolled back using log entries.
- **Redo Phase (Durability):** Committed transactions that had not yet been flushed from RAM to table storage are re-applied from log entries.

Q3: Explain Transaction Isolation Levels and the anomalies they prevent.

SQL defines four standardized isolation levels balancing performance and concurrency isolation:

Isolation Level	Dirty Read	Non-Repeatable Read	Phantom Read
Read Uncommitted	Allowed	Allowed	Allowed
Read Committed	Prevented	Allowed	Allowed
Repeatable Read	Prevented	Prevented	Allowed
Serializable	Prevented	Prevented	Prevented